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Chu Ngọc Minh Khôi - 2410478

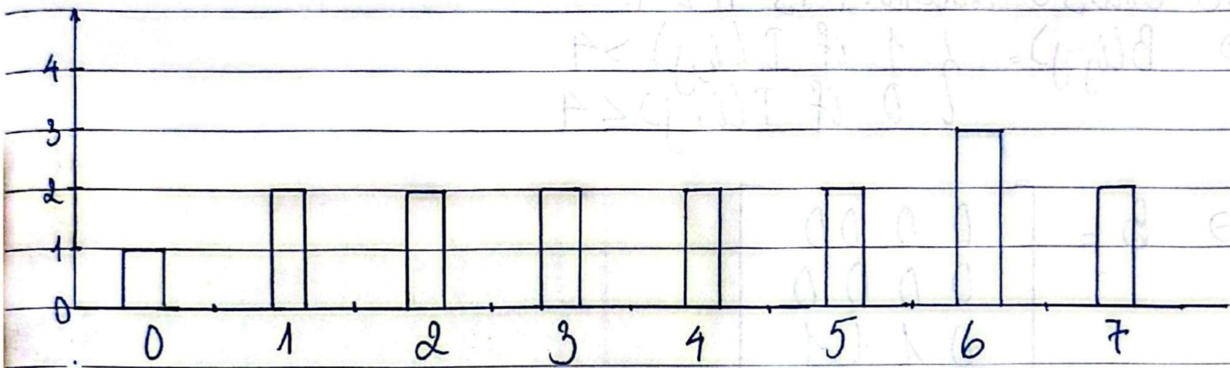
Part A:

Task A1:

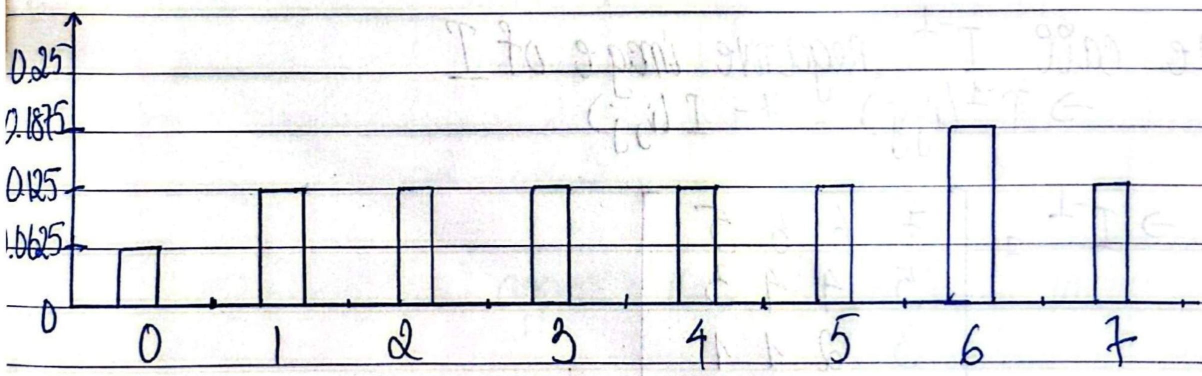
a) We have $N = 4 \times 4 = 16$

Intensity	0	1	2	3	4	5	6	7
Histogram	1	2	2	2	2	2	3	2
N. Histogram	0.0625	0.125	0.125	0.125	0.125	0.125	0.1875	0.125

Histogram



N Histogram



b) We have:

Intensity	0	1	2	3	4	5	6	7
Histogram	1	2	2	2	2	2	3	2
cdf	1	3	5	7	9	11	14	16
N. cdf	0.0625	0.1875	0.3125	0.4375	0.5625	0.6875	0.875	1
N. cdf * L	0.4375	1.3125	2.1875	3.0625	3.9375	4.8125	6.125	7
New Intensity	0	1	2	3	4	5	6	7

(L : intensity max = 7)

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We see that the new intensity is same with initial intensity, so we can have:

$$I4 = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 3 & 3 & 4 \\ 4 & 5 & 6 & 6 \\ 5 & 6 & 7 & 7 \end{bmatrix}$$

c) We have all pixel value sorted:

0, 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6, 6, 7, 7

We choose threshold is $h = 4$.

$$\Rightarrow B(i,j) = \begin{cases} 1 & \text{if } I(i,j) > 4 \\ 0 & \text{if } I(i,j) \le 4 \end{cases}$$

$$\Rightarrow B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

d) We call I^{-1}: negative image of I

$$\Rightarrow I^{-1}(i,j) = 7 - I(i,j)$$

$$\Rightarrow I^{-1} = \begin{bmatrix} 7 & 6 & 6 & 5 \\ 5 & 4 & 4 & 3 \\ 3 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

Task A2:

a) We have $N = 5 \times 5 = 25$ pixels.

Intensity	0	1	2	3	4	5	6	7
Histogram	2	2	6	7	2	1	3	2
N. Histogram	0.08	0.08	0.24	0.28	0.08	0.04	0.12	0.08

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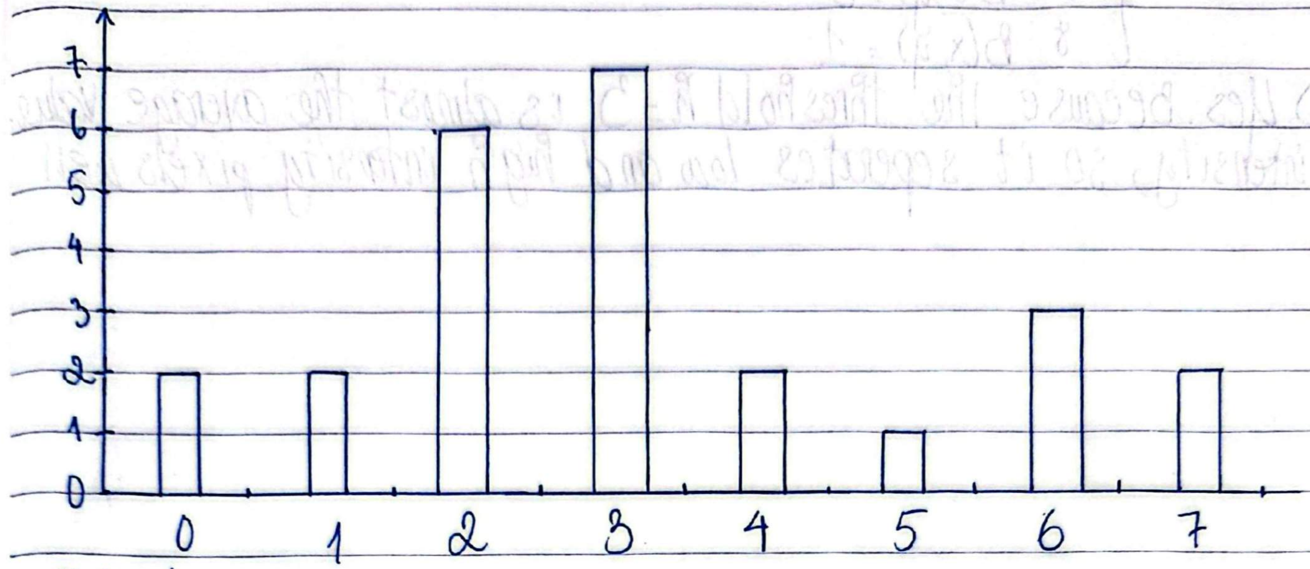
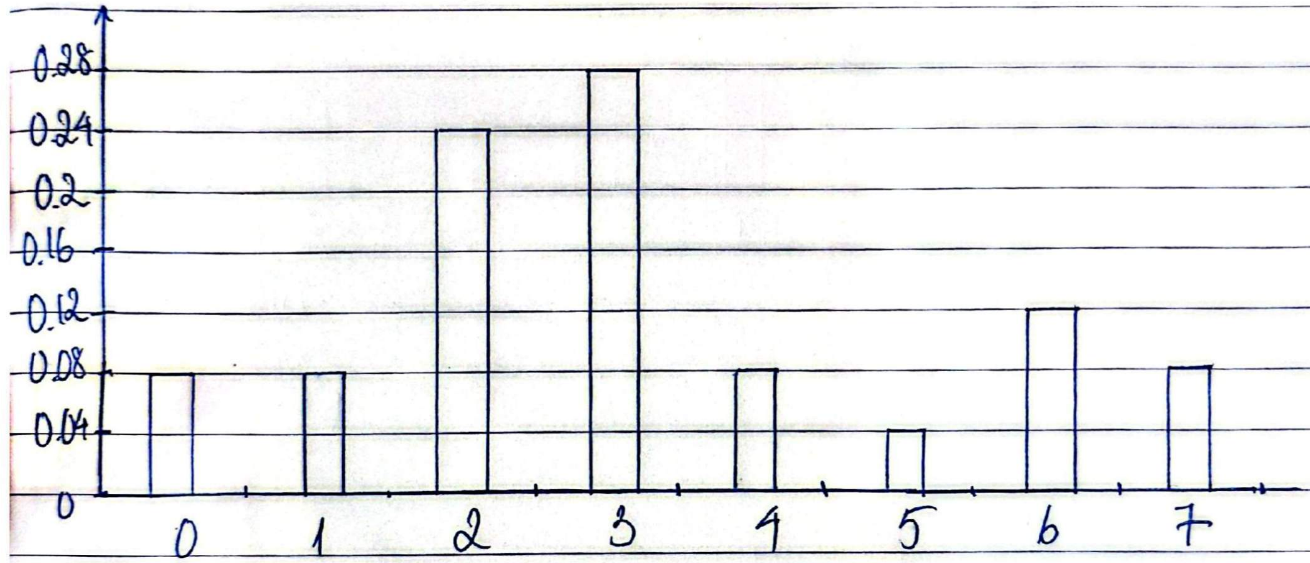
Histogram



N. Histogram



b) The gray-level that appears most frequently in image J is intensity 3

c) By question b), we chose threshold $h = 3$

$$\rightarrow B(i,j) = \begin{cases} 1 \text{ if } J(i,j) > 3 \\ 0 \text{ if } J(i,j) \leq 3 \end{cases}$$

$$\rightarrow B = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

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d) By question c) we have:

$$\begin{cases} 17 & B(x,y) = 0 \\ 8 & B(x,y) = 1 \end{cases}$$

e) Yes. Because the threshold $k = 3$ is almost the average value of intensity, so it seperates low and high intensity pixels well